

Data Partitioning and Pipelining in Parallel Execution

1. Data Partitioning

What it means:

Splitting big data into smaller parts and sending them to different computers to process at the same time.

Problem: Uneven Data (Data Skew)

Example:

- Computer 1 gets 10,000 records
- Computer 2 gets 500 records
- Computer 3 gets 300 records

Problem: Computer 1 is very slow, others finish early and stay idle.

Solution:

Split the data evenly using a method like hashing, so all computers get similar amounts of work.

2. Pipelining

What it means:

Doing tasks in stages, where one task gives output to the next task while running together.

Problem: Blocking Operations

Example:

`SELECT AVG(Marks) FROM Student;`

- First: Collect all marks
- Then: Calculate average

Problem: The average cannot be calculated until all data is collected.

Solution:

- Calculate partial averages at each computer
- Then combine them at the end.

Summary Table

Data Partitioning and Pipelining in Parallel Execution

Problem Type	Simple Example	Issue	Solution
Data Partitioning	One computer gets more data than others	Slow performance, waste of resources	Split data evenly (e.g., hashing)
Pipelining	AVG(Marks) needs all data first	Later stages must wait (blocking)	Do partial work first, combine later